*Proyecto 1.

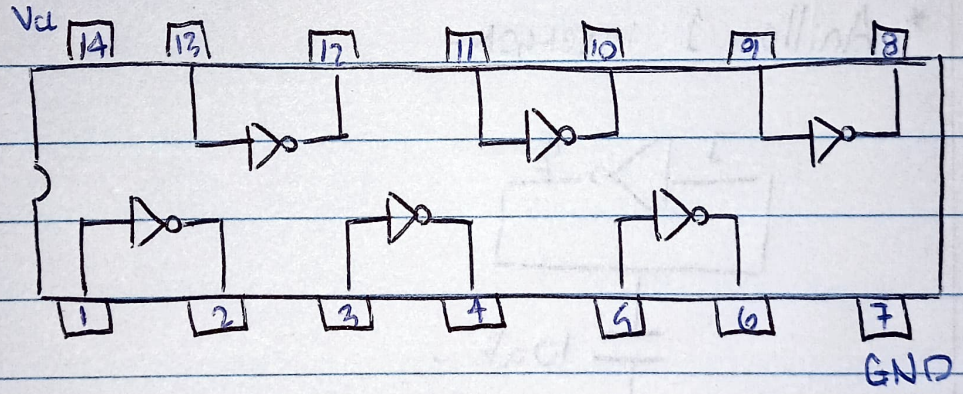
Ejercicio 2. Oscilador de Anillo.

Fórmulas para el cálculo del tiempo de propagación del inversor TTL.

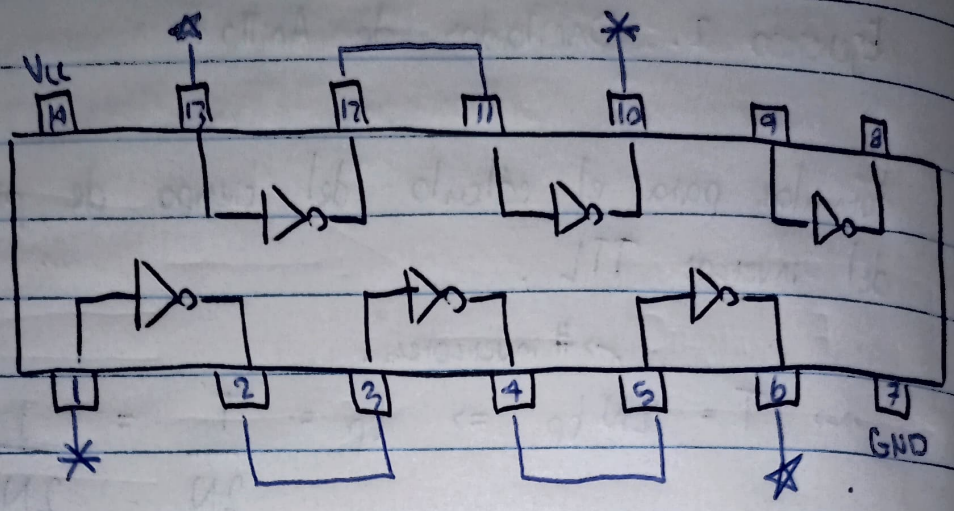
$$\rightsquigarrow T = \underbrace{2N}_{\#\text{ inversores.}} t_p \Rightarrow t_p = \frac{T}{2N} = \frac{1}{2Nf}$$

$$\rightsquigarrow t_p = \frac{t_{LH} + t_{HL}}{2}$$

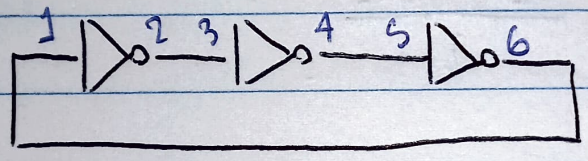
* Hex Inverter.



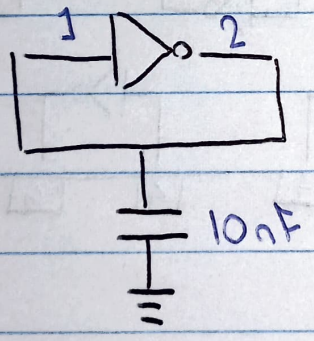
* Anillo 5 inversores



* Anillo 3 inversores.



* Anillo 1 inversor.



* Cálculos Laboratorio

5 Anillo. $t_{PD} = 3{,}3898\,ns$

t_{PD} : $x_1 = -42\,ns$ $x_2 = -40\,ns$

3 Anillo. $t_{PD} = 5{,}005\,ns$

t_{PD} : $x_1 = -8{,}1\,ns$ $x_2 = -6\,ns$

$Máx_1 = 2{,}31\,V$ $mín_1 = 770\,mV$ $Máx_2 = 1{,}87\,V$ $mín_2 = 440\,mV$.

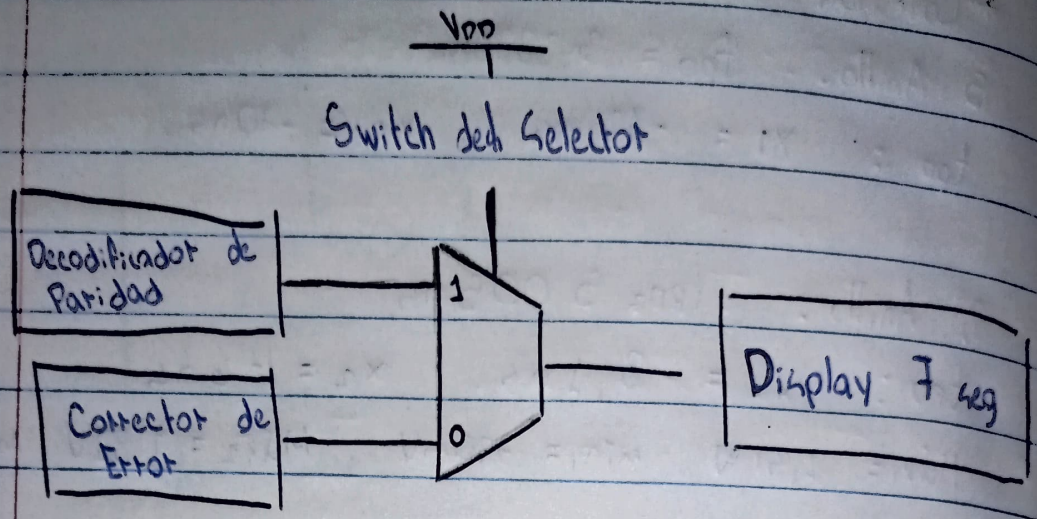
con 1 m de Alambre:

t_{PD} $x_1 = -20\,ns$ $x_2 = -27\,ns$

1 Anillo:

Con C: 1,01 V y Sin : 930 mV

- Submódulo Selector.



→ Ecuaciones booleanas.

Entrada 1 : $D = (D_3, D_2, D_1, D_0)$ → Palabra Corregida

Entrada 2: $S = (S_3, S_2, S_1, S_0)$ → Posición.

Control : selec.

$$\rightsquigarrow \quad Y_i = \overline{selec} \cdot D_i + selec \cdot S_i$$